P4 Autocomplete — Analysis

Question 1 — Benchmark results

BenchmarkForAutocomplete run with all three implementations on each file.

threeletterwords.txt

```
init time: 0.009105    for BruteAutocomplete
init time: 0.003237    for BinarySearchAutocomplete
init time: 0.04811    for HashListAutocomplete
search    size    #match    BruteAutoc    BinarySear    HashListAu
    17576    50    0.00268167    0.00491604    0.00001175
    17576    50    0.00063450    0.00332383    0.00000671
a    676    50    0.00055113    0.00028188    0.00001479
a    676    50    0.00052646    0.00027338    0.00000392
b    676    50    0.00052833    0.00028154    0.00000379
c    676    50    0.00067133    0.00023217    0.00000567
g    676    50    0.00044950    0.00019988    0.00000425
ga    26    50    0.00027304    0.00003225    0.00000388
go    26    50    0.00017183    0.00002446    0.00000308
gu    26    50    0.00017950    0.00002754    0.00000367
x    676    50    0.00021913    0.00020329    0.00000442
y    676    50    0.00021988    0.00019492    0.00000404
z    676    50    0.00021196    0.00019683    0.00000375
aa    26    50    0.00017250    0.00002450    0.00000354
az    26    50    0.00017221    0.00002408    0.00000342
za    26    50    0.00017467    0.00002538    0.00000329
zz    26    50    0.00017200    0.00002467    0.00000350
zqzqwwx    0    50    0.00013333    0.00001525    0.00004500
size in bytes=246064     for BruteAutocomplete
size in bytes=246064     for BinarySearchAutocomplete
size in bytes=1092468     for HashListAutocomplete
```

fourletterwords.txt

```
init time: 0.04516    for BruteAutocomplete
init time: 0.02410    for BinarySearchAutocomplete
init time: 0.5947    for HashListAutocomplete
search    size    #match    BruteAutoc    BinarySear    HashListAu
    456976    50    0.00632529    0.03406238    0.00001221
    456976    50    0.00338767    0.00341071    0.00001125
a    17576    50    0.00402129    0.00223746    0.00000825
a    17576    50    0.00342942    0.00114725    0.00001263
b    17576    50    0.00259038    0.00060079    0.00000950
c    17576    50    0.00248767    0.00053725    0.00000950
g    17576    50    0.00253946    0.00047063    0.00001504
ga    676    50    0.00243479    0.00010429    0.00000842
go    676    50    0.00245050    0.00007246    0.00001008
gu    676    50    0.00251096    0.00008004    0.00001021
x    17576    50    0.00238488    0.00039121    0.00000904
y    17576    50    0.00242592    0.00028700    0.00001133
z    17576    50    0.00244508    0.00024950    0.00000963
aa    676    50    0.00247567    0.00007254    0.00000933
az    676    50    0.00253079    0.00006854    0.00000867
za    676    50    0.00231779    0.00007208    0.00000842
zz    676    50    0.00236450    0.00007546    0.00000950
zqzqwwx    0    50    0.00247317    0.00006946    0.00005338
size in bytes=7311616     for BruteAutocomplete
size in bytes=7311616     for BinarySearchAutocomplete
size in bytes=40322100     for HashListAutocomplete
```

alexa.txt

```
init time: 0.1520    for BruteAutocomplete
init time: 0.8787    for BinarySearchAutocomplete
init time: 4.082    for HashListAutocomplete
search    size    #match    BruteAutoc    BinarySear    HashListAu
    1000000    50    0.01441146    0.05872142    0.00001871
    1000000    50    0.00620642    0.03362825    0.00003458
a    69464    50    0.00770054    0.01394863    0.00002217
a    69464    50    0.00840046    0.00758175    0.00002483
b    56037    50    0.00710267    0.00376750    0.00001392
c    65842    50    0.00732088    0.00342863    0.00001338
g    37792    50    0.00674558    0.00191858    0.00001358
ga    6664    50    0.00904200    0.00059996    0.00002488
go    6953    50    0.00694529    0.00057871    0.00001383
gu    2782    50    0.00683721    0.00030604    0.00001404
x    6717    50    0.00643929    0.00051988    0.00001442
y    16765    50    0.00635125    0.00101363    0.00002521
z    8780    50    0.00524404    0.00055925    0.00001233
aa    718    50    0.00733396    0.00014013    0.00001317
az    889    50    0.00838521    0.00034658    0.00006629
```

```
za    1718    50    0.00613283    0.00022713    0.00001367
zz    162    50    0.00606508    0.00010896    0.00002538
zqzqwwx    0    50    0.00631354    0.00084708    0.00036392
size in bytes=38204230     for BruteAutocomplete
size in bytes=38204230     for BinarySearchAutocomplete
size in bytes=475893648     for HashListAutocomplete
```

Question 2 — Brute vs. BinarySearch on the empty prefix

The empty prefix matches every term, so M = N. BinarySearch's only advantage is locating the matching range in O(log N) instead of scanning the whole array. When M = N that range *is* the whole array, so both implementations still do the same O(M log k) = O(N log k) work over every term. BinarySearch even does a little extra (the two binary searches plus building the matching sublist), so Brute ties it or is slightly faster. This is a real effect from M = N, not just small-clock noise.

Question 3 — alexa.txt with matchSize (k) = 100

```
init time: 0.1505    for BruteAutocomplete
init time: 0.8475    for BinarySearchAutocomplete
init time: 3.387    for HashListAutocomplete
search    size    #match    BruteAutoc    BinarySear    HashListAu
    1000000    100    0.02096250    0.07282292    0.00002254
    1000000    100    0.00696229    0.03594783    0.00003858
a    69464    100    0.00883792    0.01589129    0.00001738
a    69464    100    0.00751292    0.01180738    0.00011175
b    56037    100    0.05363063    0.00964029    0.00008292
c    65842    100    0.00783883    0.00414279    0.00007758
g    37792    100    0.00703308    0.00222008    0.00001463
ga    6664    100    0.00696271    0.00066608    0.00001321
go    6953    100    0.00711067    0.00067900    0.00001438
gu    2782    100    0.00692271    0.00036504    0.00001308
x    6717    100    0.00640963    0.00063417    0.00001454
y    16765    100    0.00669092    0.00108833    0.00001450
z    8780    100    0.00656438    0.00069729    0.00001446
aa    718    100    0.00974454    0.00028842    0.00013800
az    889    100    0.00857200    0.00021058    0.00001746
za    1718    100    0.00784017    0.00048633    0.00010975
zz    162    100    0.00691850    0.00024513    0.00003383
zqzqwwx    0    100    0.00655163    0.00014683    0.00030058
size in bytes=38204230     for BruteAutocomplete
size in bytes=38204230     for BinarySearchAutocomplete
size in bytes=475893648     for HashListAutocomplete
```

Yes, the data support logarithmic dependence on k. Doubling k from 50 to 100 barely changed the Brute and BinarySearch times (they came nowhere near doubling). Since the per-query cost grows as M log(k), doubling k only multiplies log(k) by about log(100)/log(50) ≈ 1.18 (~18%), not 2x. The small increases we see match that logarithmic growth rather than the 2x a linear dependence on k would produce.

Question 4 — Why HashList is fast but memory-heavy

HashListAutocomplete precomputes, for every prefix up to MAX_PREFIX, the already weight-sorted list of matching terms. So topMatches is just a HashMap lookup plus a sublist — effectively O(1), with no scanning or sorting at query time. The tradeoff is memory: each term is stored in up to MAX_PREFIX+1 prefix lists, plus every prefix string is kept as a key, so storage is many times the single Term array the other two use. sizeInBytes shows this: 1,092,468 vs 246,064 bytes on threeletterwords, and 475,893,648 vs 38,204,230 bytes on alexa.